# ECE501 - DIGITAL IMAGE PROCESSING

## SAR-Optical Feature Fusion for Crop-Type Classification

## REPORT - WEEK 5


| Name | Enrollment No. |
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**GitHub Repository:**
https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

**1) Work done this week**

→ Change of data source. The preprocessed YieldSAT files store field locations only as encoded coord_x, coord_y and coord_z values, which could not be converted back to latitude and longitude, so Sentinel-1 data could not be matched to fields. We moved to the YieldSAT raw per-field release (original-preprocessed). In this release every field has its own folder with a metadata JSON file, Sentinel-2 images, SCL cloud masks, DEM, soil, weather and yield masks.

→ Raw-data processing pipeline defined: Defined the processing steps for the raw data: (1) read the raw JSON files and build an inventory file for Germany and Uruguay; (2) extract the period from sowing date to harvest date for each field; (3)

download the matching Sentinel-1 SAR data locally as one file per field of about 5–10 MB; (4) co-register the SAR data to the 10 m optical grid; and (5) verify the SAR data against the raw JSON field geometry in QGIS by checking the overlap between the SAR tile and the raw field data.

**2) Work Planned for Next Week**

1. Inspect the raw JSON files for Germany and Uruguay and build the inventory file (field id, crop, sowing date, harvest date, location).
2. Download Sentinel-1 data for a few fields first, check file sizes against the 5–10 MB target, then download for all fields.
3. Check co-registration and verify the SAR data against the raw JSON geometry in QGIS, recording the overlap results.
4. Run the optical feature extraction on the Germany data and save the feature table and plots to the repository.
5. Start Sentinel-1 feature extraction (VV and VH statistics, VV/VH ratio, texture) once verification is complete, and report fusion or model-accuracy results only after they have been experimentally verified.

3) **Preliminary Results and Observations**

| **Country** | **Crop(s)** | **Fields** |
|---|---|---|
| Germany | Rapeseed, wheat | 299 (111 rapeseed, 188 wheat) |
| Uruguay | Soybean | 572 |

The raw data for the two countries is very large, so building the inventory and downloading Sentinel-1 data for every field takes substantial processing time. This is the main constraint on the pipeline at present.

**4) Challenges Faced**

→ Data volume. The raw release contains many Sentinel-2 images and layers per field and is much larger than the preprocessed files, so we work through the JSON metadata and a small batch of fields first.

→ Field extent. The JSON provides the centroid and area but not the field polygon. The export region is therefore a square around the centroid with a margin, and the exact extent has to be taken from the Sentinel-2 raster when co-registering.

→ Earth engine limits. The free tier has a limited compute quote and a limit on export tasks, and the file size depends on the field area the number of Sentinel-1 acquisition in the season, so the 5-10 MB target has to be checked per field.

**5) References:**

→ Google Earth Engine – Sentinel-1 SAR GRD.
Documentation describing the Sentinel-1 Ground Range Detected dataset and its available radar polarization bands.
https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD

→ Google Earth Engine – Sentinel-2 Surface Reflectance Harmonized.
Documentation describing Sentinel-2 surface reflectance data and the available optical bands.
https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

→ A. Miranda et al., "YieldSAT: A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction," CVPR 2026. Project page: https://yieldsat.github.io/

→ Studies on Sentinel-1/Sentinel-2 fusion for crop-type mapping in Germany and South America